

003 — Launch Cost Curve & Value Migration

Launch cost is the sector's master cost variable and not its master value variable. The curve is real but thin — one vehicle has a measured marginal cost, and on the corrected denominator its cost per kilogram rose 32.0% while its cost per launch fell 12.0%. The value pool migrated to whoever owns the demand, and the released value did not pass through to the payload customer.

Constitution pin	1.5.0
As of	2026-09-19
Universe	SPCX 20% · RKLБ 20% · FLY 10% · SATS 8% · IRDM 8% · GSAT 7% · LUNR 7% · PL 10% · YSS 10%
Generated	2026-09-19 12:06 UTC

1. The cost curve's only measured point runs backwards
2. One vehicle discloses a marginal cost, and it is the only one
3. Cost per launch fell while cost per kilogram rose
4. The sector's own scoreboard sets every cost denominator
5. The denominator, not the measurement, is the whole argument
6. Four correct quotients span twelve point eight times
7. The value pool moved to whoever owns the demand
8. Where the curve is claimed and nowhere demonstrated
9. Neutron closes on one basis and fires on another
10. Three of four demand-side names cannot draw the number
11. The register that classifies defects was defeated by one
12. Five corrections this thesis made to its predecessors
13. What 004 inherits, and the case against it



The cost curve's only measured point runs backwards

+32.0% ELECTRON COST PER KG the curve's one measured vehicle, corrected basis	12.8× SPREAD, ONE PAGE four correct \$/kg values, zero basis fields	1 of 5 VEHICLES WITH A COSTED POINT the other four rest on claims or models	9 of 9 TICKERS SIGN- STRIPPED including the analysis written to document it
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The news: the sector's cost-curve story rests on **one vehicle** that discloses its marginal cost. Measured against the right denominator, that single point goes **the wrong way** — cost per launch fell **12.0%** while cost per **kilogram rose 32.0%**. The two numbers are the same fact.

Why that is not a rounding argument. Electron flew six missions in Q2 2026 and **two of them were suborbital testbeds carrying zero kilograms to orbit**. Cost per launch divides by six; cost per kilogram divides by four. $0.88 \times 1.50 = 1.32$ exactly — the inversion is arithmetic, not interpretation.

What this means for the programme. Any thesis pricing a launch-cost decline into a revenue model is pricing a **number that has not appeared**. The decline is real in the denominator everyone quotes and **absent in the denominator that matters**. That is the widest-reaching result here.

The second finding is about ambiguity, and it is five times larger. SPCX's 8-K prints four numerators and two denominators **on one page**; the four cost-per-kilogram values that follow span **12.8×** and **none of them is a calculation error**. The spread decomposes exactly — numerator **1.617×**, denominator **7.886×**.

What it does not mean. The value pool did not vanish, it **moved**. Two acquirers bought the customer relationship underneath two independent operators within twelve months, and **neither paid for the margin**. Anchor (a) holds; anchor (b) is **falsified**.

How to read this report. The one measured point comes first, then the denominator argument that generalises it, then where the money actually went, then the register — **which was itself defeated by one of the defects it classifies**, inside an artifact written to document it, with all three automated gates passing. **The recommendation is not a trade. This thesis produces no positions by design. It produces a citation rule: carry the basis with the grade.**

THE CURVE · ONE VEHICLE, EIGHT PERIODS

One vehicle discloses a marginal cost, and it is the only one

The 20-cell matrix crosses **five vehicles** against **four cost bases**. Two cells are DEMONSTRATED, and **both belong to Electron**. Thirteen cells are **ABSENT** — not unmeasured by this thesis, but **not disclosed by anyone**.

Vehicle	Issuer	Architecture	Best populated basis	Grade
Electron	RKLB	fully expendable	basis B — marginal cost per launch	DEMONSTRATED
Falcon 9	SPCX	partially reusable	basis A — list price	CLAIMED
Starship	SPCX	declared F5a · as-flown F5c	basis C — amortized	CLAIMED
Neutron	RKLB	partially reusable, unflown	basis A — list price	MODELED
Alpha	FLY	fully expendable	basis A — list price	CLAIMED

Basis B is the only basis that tests an architecture floor, because it is the only one that reports what a launch *costs* rather than what it is *sold for*. Electron is the only vehicle in the universe where it is filed. **A demonstrated cost curve with a population of one is a point, not a curve** — and this thesis reports it as a point.

What this means. The sector's productivity narrative is **not falsified here**; it is **unfalsifiable from the disclosures as they stand**. Four of five vehicles cannot be tested at all, which is a different and harder problem than four of five failing.

THE INVERSION · COST PER LAUNCH AGAINST COST PER KILOGRAM

Cost per launch fell while cost per kilogram rose

Same vehicle, same quarter, same filings. One denominator moved down and the other moved up, and the gap between them is the HASTE mission mix.

Measure	Q2 2025	Q2 2026	Change	What divides what
Cost per launch	\$5.0M	\$4.4M	-12.0%	cost ÷ 6 launches
Cost per kilogram to LEO	\$16,667	\$22,000	+32.0%	cost ÷ 4 orbital launches
HASTE suborbital missions	0	2	—	carry zero kg to LEO

The identity is exact. $0.88 \times 1.50 = 1.32$. Six missions to four is a factor of **1.50**; the cost line fell to **0.88** of its prior level; the product is the **1.32** increase in cost per kilogram. **Nothing here is estimated.** Both the cost line and the launch counts are filed — [RKL B 10-Q p.6](#).

Why the correction is not optional. A suborbital testbed that reaches no orbit delivers **no kilograms**. Counting it in the denominator of a *cost-per-kilogram* metric makes the metric move for a reason that has nothing to do with cost per kilogram. **The uncorrected series would have shown a decline.**

What this means. If the sector's productivity claim is true, **this is the one place it should be visible**, and it is not. That does not prove the claim false for Starship or Neutron — **it proves the claim has never been tested where it is testable**, which is the narrower and more uncomfortable finding.

THE SCALE · DENOMINATORS BEFORE COST

The sector's own scoreboard sets every cost denominator

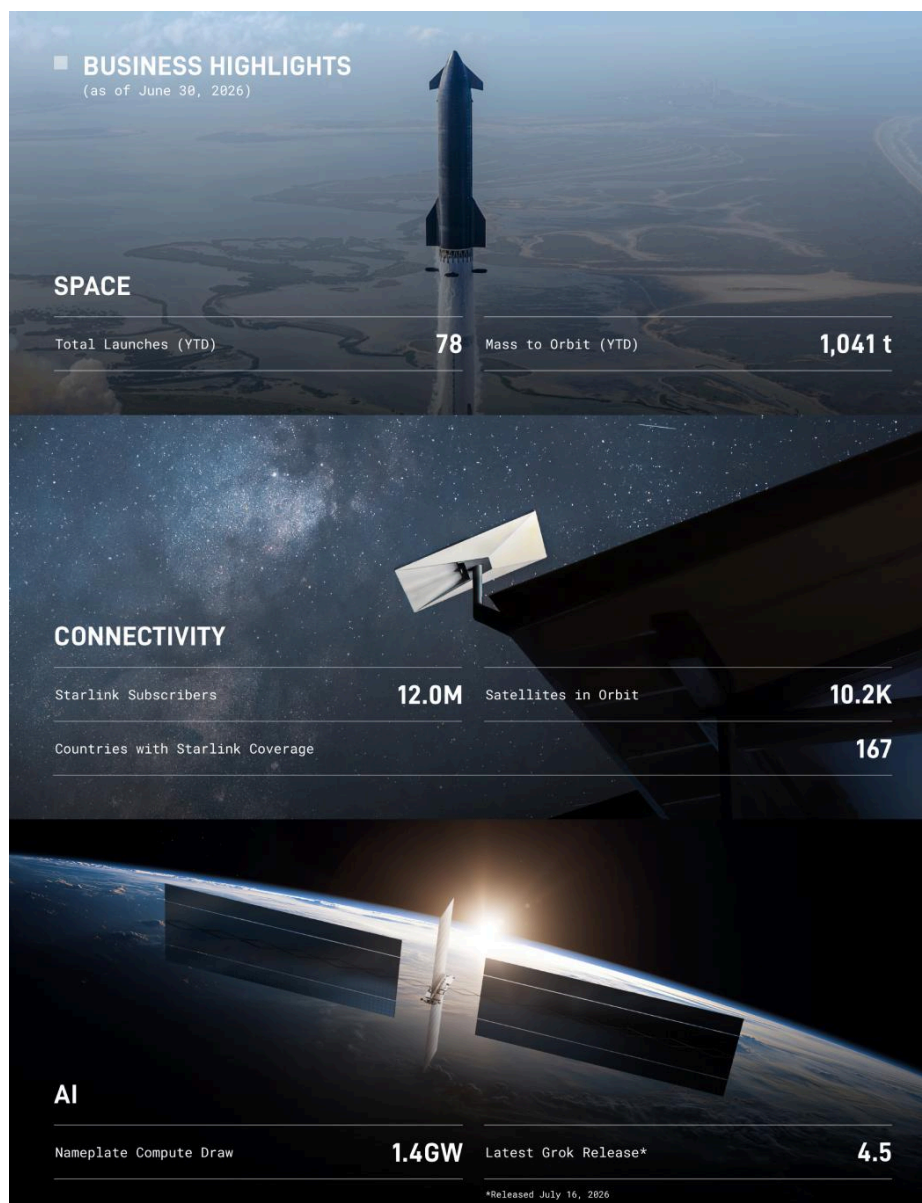


Exhibit 99.1, printed page 2, at reference scale and unretouched — the one page in this filing set carrying imagery rather than tables. **78 launches** and **1,041 metric tons** to orbit, six months ended 2026-06-30, are the two denominators under every cost-per-kilogram figure here; **12.0M** subscribers and **1.4GW** of compute are the demand and enabling ends. Issuer-produced and unaudited. Source: [SPCX 8-K p.5](#)

What this means for the pages that follow. These are the numbers the ladder divides by, so the sector's **scale is not in dispute** — the count and the tonnage are filed, and this thesis does not correct them. **What is in dispute is which belongs underneath a cost figure**, and that choice moves the answer by 12.8×

THE DENOMINATOR · ONE PAGE, FOUR CORRECT VALUES

The denominator, not the measurement, is the whole argument

Space Segment Operating and Financial Data

\$, in millions (except where noted)	Three Months Ended			Six Months Ended	
	June 30, 2026	March 31, 2026	June 30, 2025	June 30, 2026	June 30, 2025
Customer launches (#)	10	7	9	17	21
Internal launches (#)	28	33	37	61	63
Total launches (#)	38	40	46	78	84
Customer payloads (in metric tons)	87	45	88	132	163
Internal payloads (in metric tons)	397	511	563	908	938
Mass to orbit (in metric tons)	485	556	652	1,041	1,102
Launch services revenues	\$648	\$330	\$490	\$978	\$1,056
Launch & development revenues	314	289	256	603	555
Space revenues	\$962	\$619	\$746	\$1,581	\$1,611
Less:					
Cost of revenue	\$329	\$281	\$330	\$610	\$627
Research and development	1,076	930	693	2,006	1,219
Selling, general, and administrative	99	70	87	169	175
Impairment	—	—	5	—	29
Loss from operations	\$(542)	\$(662)	\$(369)	\$(1,204)	\$(439)
Segment Adjusted EBITDA reconciliation					
Loss from operations	\$(542)	\$(662)	\$(369)	\$(1,204)	\$(439)
Depreciation & amortization	158	166	146	324	308
Share-based compensation	179	145	125	324	233
Impairment	—	—	5	—	29
Space Adjusted EBITDA	\$(205)	\$(351)	\$(93)	\$(556)	\$131
Space Capex	\$1,174	\$1,052	\$946	\$2,226	\$1,705

Space Business Highlights

- Space revenues grew 55% sequentially and 29% year-over-year to \$962 million, driven by a higher number of large customer launches and a favorable customer shift compared to the prior year
- Total costs and expenses for the Space segment were up by \$389 million year-over-year, as we continued to accelerate R&D investments in our Starship program, which we believe will reduce the cost to orbit by 99% or more relative to the historical average, and unlock significant revenue potential across all business segments
- Leading launch provider for the world with 78 launches and 1,041 metric tons of mass to orbit deployed over the six months ended June 30, 2026, primarily allocated to Connectivity for the deployment of our Starlink constellation
- Starship V3 development continued to advance towards full and rapid reusability:
 - Completed Starship V3's first suborbital mission in May, Flight 12, which achieved a successful lift off from our new Starbase pad, a precision landing of Starship's upper stage, and deployment of modified V2 Starlink satellites
 - Subsequent to the second quarter, completed Starship Flight 13 in July, which achieved all flight objectives including deploying 20 production V3 satellites, demonstrating in-space relight of a Raptor engine, and executing the softest ever splashdown of Starship, providing critical views of an intact heatshield



4

Exhibit 99.1, printed page 4, reproduced at reference scale and unretouched — **the figure records what the issuer printed, and its citation link is the legible copy**. Launch services revenue **\$978M**, Space segment revenue **\$1,581M**, mass to orbit **1,041 metric tons**, customer payload **132 metric tons** — all filed here, six months ended 2026-06-30. The four \$/kg values the sector quotes are **derived from these, not printed on them**, which means any single figure quoted from this page **carries a basis its quoter chose and did not state**. Issuer-produced and unaudited. Source: [SPCX 8-K p.7](#)

THE LADDER · NINE RUNGS, EVERY BASIS NAMED

Four correct quotients span twelve point eight times

Same page, same six months, **four defensible numerators over two defensible denominators**. Every one is arithmetically correct. **None is comparable to any other.**

Numerator (basis)	Denominator (basis)	\$/kg	Correct?	Comparable?
Launch services revenue \$978M	Customer payload 132 t	\$7,409	yes	—
Space segment revenue \$1,581M	Customer payload 132 t	\$11,977	yes	no
Launch services revenue \$978M	Mass to orbit 1,041 t	\$939	yes	no
Space segment revenue \$1,581M	Mass to orbit 1,041 t	\$1,519	yes	no

The spread is almost entirely denominator. Numerator factor $1,581 / 978 = 1.617\times$; denominator factor $1,041 / 132 = 7.886\times$; product $12.75\times$. **The ambiguity is five times larger on the axis nobody argues about.**

The boundary that makes the numerator ambiguous is filed, and it is the customer boundary. Space segment revenue "*only reflects our customer launches and customer activities*" [SPCX 10-Q p.35](#), and for internal Starlink launches "*the Company does not recognize any inter-segment revenue*" [SPCX 10-Q p.36](#). So the numerator **excludes the majority of launches** while the denominator **includes them** — **73.0%** of launches were internal. **Putting those two on one ratio is not a measurement.**

What this means. The spread will be read as *uncertainty* when it is actually *definition*. A reader who treats it as noise will **average four values that are not commensurable** — which is why the ladder, not a single number, is the deliverable.

THE VALUE POOL · WHERE THE MARGIN SITS, AND WHAT BUYERS PAID FOR IT

The value pool moved to whoever owns the demand

Sixteen entries across the universe, graded. The pattern is **bimodal** — high at the demand-owning end, negative through the middle, and **one company holds both ends at once**. **SPCX is first of three segments on gross margin and last of three on operating**: the segment earning the highest gross margin loses the most money per revenue dollar.

Name	Position	Operating margin	Grade
SPCX Connectivity	owns the subscriber	+38.59%	DEMONSTRATED
IRDM	owns the subscriber	+15.10%	DEMONSTRATED
SATS	spectrum holder	+10.71%	DEMONSTRATED
GSAT	owns the subscriber	-7.37%	DEMONSTRATED
YSS	spacecraft prime	-44.64%	DEMONSTRATED
SPCX Space	launch + spacecraft	-56.34%	DEMONSTRATED
FLY	launch manufacturer	-80.90%	DEMONSTRATED

Two acquirers then paid for that position, twelve months apart — and neither paid for the margin.

	GSAT	IRDM
Acquirer	Amazon	RKLB
Consideration per share	\$90.00	\$54.00
Operating margin	-7.37%	+15.10%
Multiple	≈41.4× TTM revenue	≈8.3× TTM revenue

What this means. Margin is a property of **position**, not of sector membership — a supplier here earns **2.02×** the primes it supplies, the reverse of the intuition that the launcher captures the value. The price evidence points the same way: a **≈5.0×** multiple ratio runs *against* a 22.5-point margin differential, so the loss-making operator was bought at five times the revenue multiple of the profitable one. **The second operator anchor is falsified — FLY's -80.90% is a manufacturer**, worse than any operator here. Treat operator margin as **non-informative about strategic value**, and read a constellation's worth as **its demand relationship** — the subscriber, the spectrum, the contract. Sources: [SPCX 10-Q p.35](#), [IRDM p.6](#), [GSAT p.5](#), [SATS p.11](#), [FLY 10-Q p.9](#).

Where the curve is claimed and nowhere demonstrated

The census, stated as a result rather than a caveat. Twenty cells; the distribution is the finding.

Grade	Cells	What it means
DEMONSTRATED	2	both Electron, both basis B
DEMONSTRATED figure / MODELED divisor	2	a filed number over an assumed denominator
CLAIMED	2	an issuer assertion with no filed cost line
MODELED	1	Neutron, unflown
ABSENT	13	not disclosed by anyone, at any grade

Thirteen of twenty cells are not "unresearched" — they are undisclosed. The distinction matters because the remedies differ: some are a **research queue**, some a **tooling queue**, some a **schema-change queue**, and some **cannot be drawn at all**.

The issuer publishes two contradicting cost claims on the same day. Starship's cost to orbit is described as reducing by *"99% or more relative to the historical average"* [SPCX 8-K p.7](#) — a CLAIMED figure with no filed cost line beneath it, and one this thesis **cannot test either way**.

What this means. The sector's most-quoted productivity statistic is **an assertion by the party that benefits from it**. That is not a reason to disbelieve it — **it is a reason not to price it**.

Neutron closes on one basis and fires on another

PIL-4 registered a threshold for Neutron's implied cost per kilogram. **The registered number was wrong, and the correction changes the verdict.**

Item	Registered	Corrected	Why
Threshold	2939	5567	denominator-failed and a basis collapsed inside its own definition
Band	—	[5567, 7448]	an Electron price over a Neutron denominator

The registered figure was an Electron price divided by a Neutron denominator — a DA-30 basis collapse **inside the definition of the test itself**. Corrected, Neutron's implied cost of **\$3,846–\$4,231/kg** sits **below the band**, so the falsifier **does not fire** on the registered bar ([RKLB 10-K p.11](#)).

But the same case fires on two other bases. Neutron is **unflown**: its cost is a model over an assumed payload, and its A' breakeven payloads are **11,851 kg at a \$50M price** and **13,035 kg at \$55M** — **both above the vehicle's stated capability**. **The case closes on the registered bar and is knife-edge on a third.**

What this means. A threshold derived from **a price and a cost** is not a threshold — it is two questions. Neutron's verdict is **conditional on a basis that has never flown**, and any model that treats its cost per kilogram as an input is **treating a target as a measurement**.

THE DEMAND SIDE · A NUMBER THAT CANNOT BE DRAWN

Three of four demand-side names cannot draw the number

PIL-6's falsifier needs launch cost as a share of **programme** cost. At three of four names the quantity **cannot be constructed from anything filed**.

Name	What is filed	Can the ratio be drawn?
YSS	cost of revenues decomposes into four components	no — and the decomposition EXCLUDES launch
PL	names launch, quantifies it nowhere	no
SATS	named agreements, no dollar amount on either side	no
LUNR	a launch-amortisation line	only on a consolidated denominator

YSS is a presence finding, not an absence. Its cost of revenues decomposes **exhaustively** into four components that **do not include launch** ([YSS 10-Q p.13](#)) — so the ratio is not missing, it is **structurally not there**. That is a stronger result than a missing line item.

The one value above the bar is a denominator event. LUNR's **14.19%** becomes **2.80–4.37%** — not because launch spending changed, but because **cost of revenues grew 174.0%** underneath it. **The ratio moved because its denominator moved.**

What this means. The demand side **does not price off the curve** — and the honest form of that claim is **"unmeasurable at three of four names"**, not "zero". **A falsifier that cannot be evaluated is not a passed falsifier**, and this thesis reports it as NON-FORMABLE rather than scoring it.

The register that classifies defects was defeated by one

Data-integrity entry DA-23 records that the platform serves a filed negative as a **positive of identical magnitude**. It is confirmed at **9 of 9** tickers in this universe.

The sharpest instance is not an issuer. It is an artifact. The GSAT analysis written *to document DA-23* was itself **reproducing it on eight figures** — including a claimed divergence that existed **only because the artifact had read a stripped value**.

All three automated gates passed on the stripped version. A clean gate result is not evidence of a clean number — because the gate and the number were computed the same way.

The pair that shows the mechanism in one line. RKLB's served operating income is **+57,514,000**. The filed figure is **-57,514** in thousands. **Same digits, opposite sign, and a scale factor that hides it.** Sold as a positive, the loss reads as a gain.

The census of the defect, by issuer. PL **33 of 33** filed operating facts stripped ([PL 10-Q p.24](#)); FLY **15 of 15**; GSAT **13 confirmed instances with 16 unexercised** ([GSAT p.27](#)). **Where the test could not run, this thesis records it as UNEXERCISED — not as clean.**

What this means. Every figure a downstream thesis inherits should carry **the basis it was read on**. A number that reproduces arithmetically can still answer **a different question than its label names** — and no check that recomputes the same way will catch it.

WHAT THIS THESIS CORRECTED · FIVE FINDINGS AGAINST 001 AND 002

Five corrections this thesis made to its predecessors

Three theses on one programme, and the third is where the first two's errors surface. **Each of these was found by re-deriving from filed cells rather than by re-reading the artifact.**

#	Was	Corrected to	Cause
1	RKLB's cost per kg 1.79× – 2.62× too low	1.00× – 1.50×	that arithmetic was Falcon 9's from SPCX's table
2	PIL-4 threshold 2939	5567	a basis collapsed inside the test's own definition
3	PIL-5 axis timing	period normalisation	the gap persists at the zero-HASTE control
4	DA-26 " 19 of 19, universal "	"20 tested, 19 exhibiting"	FLY is the falsifying counterexample
5	F5a/F5b do not reach Electron	closed at v1.3.0 with F5c	the tiers are exhaustive, and the finding is an inversion

Correction 1 is the one to read twice. A cross-issuer figure was transplanted from one company's filings onto another's — **the arithmetic reproduced perfectly and belonged to a different company.** The "four of four periods" claim it supported is **falsified**: in the one period with a fully filed mix, the correction is **zero**.

What this means. A figure that reproduces is not thereby a figure about the thing you think it is. Every correction here was live in a published artifact, and **none was caught by a gate.**

ACROSS THE UNIVERSE · HANDOFF, AND THE CASE AGAINST IT

What 004 inherits, and the case against it

Six pillars, reported with their falsifier outcomes rather than their conclusions. Two are named below because they are the ones a downstream thesis will lean on.

Pillar	Verdict
PIL-1	HOLDS falsifier does not fire, on a population of one
PIL-2	HOLDS at SPCX; UNEXERCISED everywhere else
PIL-3	HOLDS demonstrated, not argued
PIL-4	KNIFE-EDGE closes on the registered bar; fires on two other bases
PIL-5	FALSIFIED in tolerance; holds in kind
PIL-6	NON-FORMABLE at three of four demand-side names

Four disposition classes are kept distinct because their remedies differ — a **research queue**, a **tooling queue**, a **schema-change queue** (REACHABLE-BUT-NOT-RECORDABLE: the contract has no admissible field, so **no amount of research fixes it**), and NON-FORMABLE. **The highest-value resolution is identified:** SPCX's DA-26 exposure cannot be exercised until **an FY2026 10-K exists** — one filing, and a register question closes.

The recommendation is not a trade. This thesis produces no positions by design. It produces a **citation rule: carry the basis with the grade** — because the alternative, demonstrated above twice, is inheriting a figure that reproduces and means something else.

Adversarial review — six findings, run against this thesis

- HIGH C-1** — the scan's zero is **vacuous, not clean: none of 45** artifacts declare entity_claims, a field the contract does not define (**0 of 128** across 001–003)
- HIGH C-2** — the inversion is **knife-edge on two integers**: reclassify one mission and **+32.0% becomes +5.6%**
- HIGH C-3** — it **names the winner and declines to act**; no position means no falsifiable conviction
- MED C-4** — "margin is non-informative" rests on **n = 2**, and the two deals have different logics
- MED C-5** — every falsifier reads a source this thesis calls **defective at 9 of 9** (DA-23)
- MED C-6** — **two of six wrong_if are not evaluable**: PIL-4's threshold moved, PIL-6 is non-formable

What this means. The recommendation above survives as a **research note, not a position.**

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